1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L30 - System Integration & Testing**

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4 **Recap**

- Learnt about
  - System implementation
- Involves
  - Code writing (follow good styles)
  - Testing (at unit level)
    - Review-based – walkthrough & inspection
    - Execution-based - black-box & white-box

5 **Recap**

- This lecture
  - Putting together of units and system testing

6 **Integration & Testing**

- System consists of subsystems (modules and units)
- Testing whole system at a time difficult
- Alternative approaches required

7 **Integration & Testing**

- TWO broad approaches
  - Bottom-up testing
  - Top-down testing

8 **Bottom-up Testing**

- Each subsystem tested separately and then full system tested

9 **Bottom-up Testing**

- A subsystem may consist of many modules communicate through well-defined interfaces
  - Primary purpose is to test the interfaces
  - Both control and data interfaces are tested
  - Test cases should exercise all interfaces in all possible manners

10 **Top-down Testing**

- Testing starts with the main routine
  - After top-level 'skeleton' tested, the immediate subroutines of the 'skeleton' are

combined with it and tested

11 **Stubs & Drivers**

- Such approaches may (likely) require
  - 'Stubs' - simulate effect of lower-level routines called by the routines under test (in top-down approach)
  - 'Driver' routines – used during bottom-up testing to 'simulate' behavior of upper level modules that are not yet integrated

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Stages of Testing

13 **System Testing**

- THREE main stages
  - Alpha testing
  - Beta testing
  - Acceptance testing

14 **Alpha Testing**

- Carried out by test team within organization

15 **Beta Testing**

- Performed by a select group of 'friendly customers' (may be specially 'recruited')

16 **Acceptance Testing**

- Performed by customer

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Types of Testing

18 **System Testing**

- What is tested?
  - Functionality
  - Performance

19 **Functionality Test**

- Test software functionality w.r.t SRS document

20 **Performance Test**

- Tests non-functional requirements

21 **Some Important Performance Tests**

- Stress testing - evaluates system performance under abnormal/illegal input conditions (in short time periods)

22 **Some Important Performance Tests**

- Volume testing – tests system performance for large input

23 **Some Important Performance Tests**

- Configuration testing – done to analyze system behavior in various hardware and software configurations specified in the requirements

24 **Some Important Performance Tests**

- Compatibility testing – checks if the system interfaces properly with other systems

25 **Some Important Performance Tests**

- Regression testing – tests backward compatibility of software with older platforms/systems

26 **Some Important Performance Tests**

- Recovery testing - tests system response to faults such as loss of power, devices, services, data, and so on

27 **Some Important Performance Tests**

- Documentation testing – tests various manuals and documents created

28 **Some Important Performance Tests**

- Usability testing – empirical testing (more on it later)

29 **Reference**

- Rajib Mall (2018). Fundamentals of Software Engineering, 5th ed, PHI Learning Pvt Ltd. Chapter 10
- Roger S Pressman (2020). Software Engineering: A Practitioner's Approach, 9th ed, McGraw-Hill Education, New York, Chapter 19-21